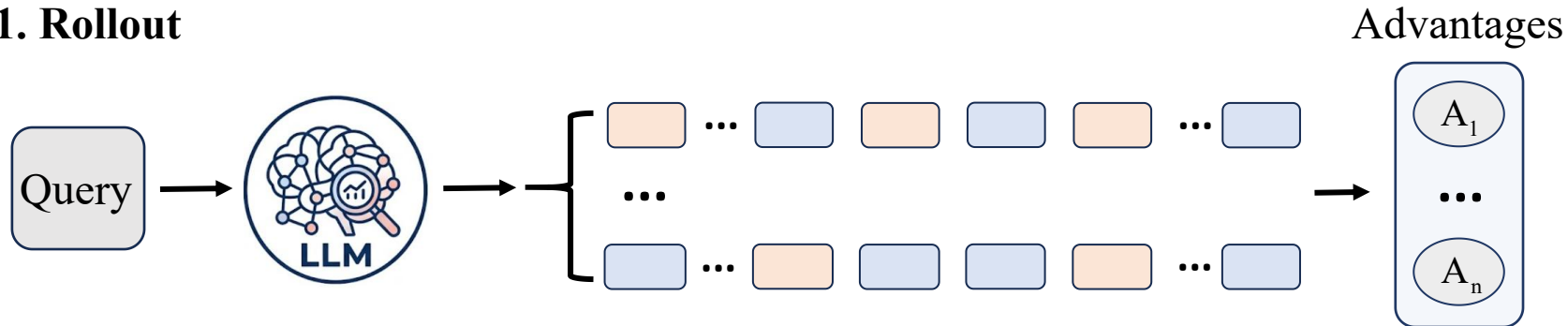


1. Rollout

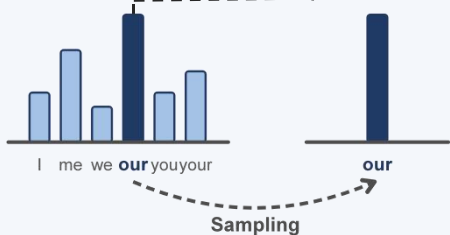


2. Action-Routing Policy Optimization

$$\mathcal{L}_{TARPO}(\theta) = \mathcal{L}_{tok} + \lambda \mathcal{L}_{act} + \beta \mathcal{L}_{KL}$$

Update Backbone

$$\mathcal{L}_{tok} = -\frac{1}{T} \sum_{t=1}^T 1_t \log \pi_{\theta}(v_{i,t} | s_{i,t}) \hat{A}_i$$



Update Action Head

$$\mathcal{L}_{act} = -\frac{1}{T} \sum_{t=1}^T \log \rho_{\theta}(d_{i,t} | h_{i,t}) \hat{A}_i$$

